YesChef

Hey chef, make me a sandwich.

A voice-controlled robot that listens, sees what's on the table, and assembles your sandwich to your liking. Just say the word and YesChef does the rest.



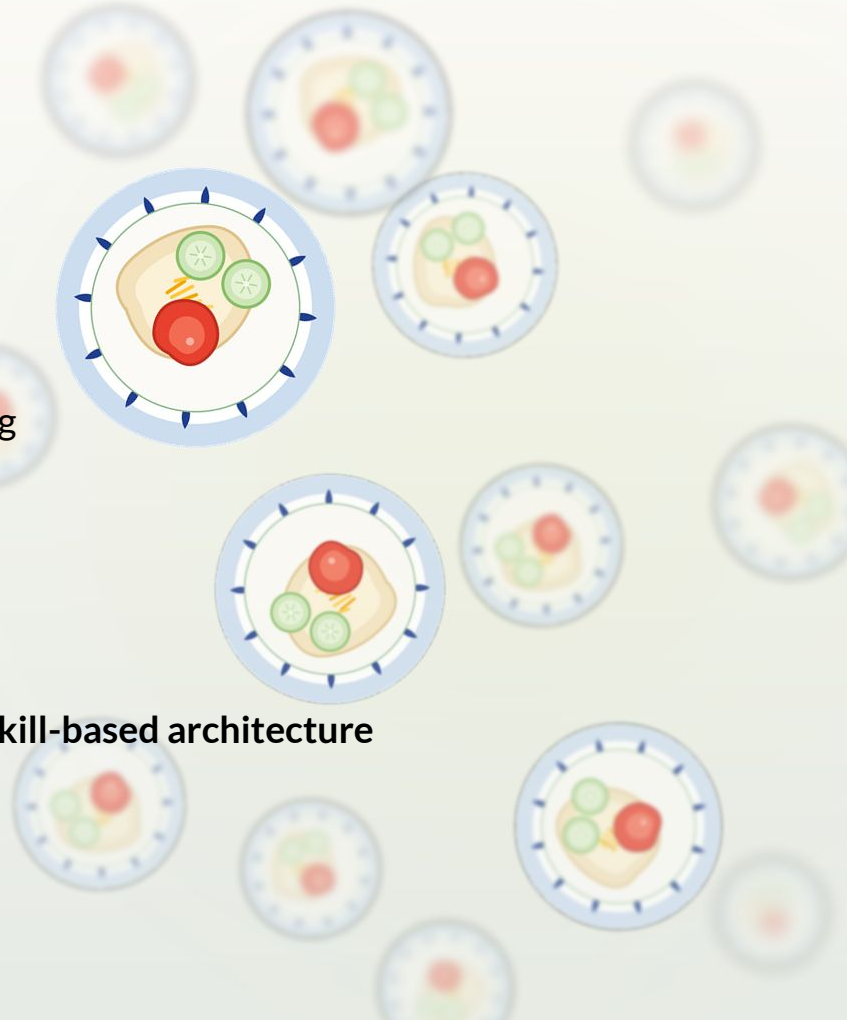
Sanja | Dhruvil | Asjad | Gautham | Kartikeya

Revolute Hackathon · Cambridge, MA · 2026

*Real Image. Non-AI Generated



Skilled Workers, One Robot



- **Make me a sandwich** is a hard robotics problem:
  - Soft, deformable ingredients defy normal grasping
  - No two ingredients are the same
  - Language has to be translated to physical action
  - Stacking is a sequential, order-dependent task
  - Has to run on an edge computer

- Our solution is **accurate & consistent** because we use **skill-based architecture**
  - Skill-based decomposition
  - VLM acts as the router
  - Each skill is fine-tuned and trajectory-optimized
  - Modular and interconfigurable

Mechanical Design

CHALLENGE

Deformable food need stable holders.

Sticky food need non-sticky solutions.

Robot has **mechanical limits** that prevent it from transferring food.

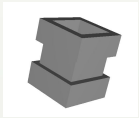
SOLUTION

Custom designed holders.
No cross-contamination.

Angled-grip for transfer feasibility.

Reinforced grip structures + non-stick layers





User Interaction

CHALLENGE

People have **food preferences, allergies** and a sequence of their sandwich content.

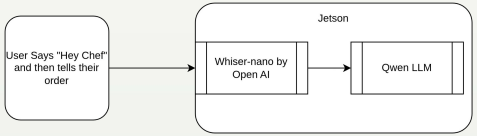
AI models don't know what is the actual order vs what is **noise**.

SOLUTION

A **speech-to-text AI model** to convert user's speech order to LLM input text.

Using a **wake word** of "Hey Chef" to discard all noise before the order.

On device (Jetson) execution of AI model for low latency



VLM Verification

CHALLENGE

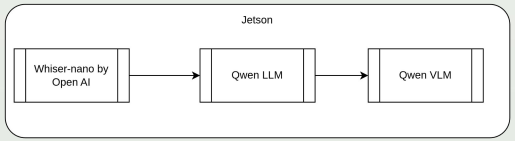
People can request items/ingredients not present on the table.

Not all VLMs can run beside an LLM on a single Jetson device accurately and comfortably.

SOLUTION

Use a VLM to understand ingredients in image of scene, and the ingredients from the Qwen LLM (that parses ingredients from audio).

Optimized VLM and LLM loading to execute ideal memory loading and latency.



Trajectory Execution / Skill Selection

CHALLENGE

It is hard for one Diffusion Policy to **learn multiple skills**.

Something needs understanding of skills available to it.

Need lots of "data" and each data point can make things more multi-modal. AKA harder to learn

SOLUTION

Do **skill-based decomposition** where we have multiple experts.

Configure the VLM to understand what skills it has access to and which ones to use when.

Collect one good trajectory and learn it really really well (over fit).

